Implementation Guidelines (V3)

1. Full Pipeline Overview

The system classifies a news post into **5 classes** by fusing three signals: **Semantic Context** (FND-CLIP), **Visual Forensics** (UnivFD), and **Textual Forensics** (Fluoroscopy).

Stage	**Component**	**Input**	**Output**	**Training Status**
Stage 1	FND-CLIP	Text + Image	v_semantic [768]	**Frozen** (Pre-trained)
Stage 2.1	DCT Preprocessing	Image File	DCT Maps [1, 224, 224]	**Math Only** (No Train)
Stage 2.2	UnivFD Encoder	DCT Map	v_imgfor [768]	**Need Training** (Stage 1)
Stage 3	Text Fluoroscopy	Text (Qwen L30)	v_textfor [768]	**Frozen** (Feature Extraction)
Stage 4	Fusion Logic	3 Vectors	Fused Vector [1024]	**Need Training** (Stage 2)
Stage 5	Classifier MLP	Fused Vector	5 Class Logits	**Need Training** (Stage 2)

2. Data Engineering & Mapping

Datasets: NewsCLIPpings, DGM4, MMFakeBench.

a. <u>A custom mapping function must unify labels into these 5 classes:</u>
b. <u>Undersampling all classes to match the smallest class's size.</u>

Index	**Label**	**Condition / Source**
0	**Real**	NewsCLIPpings (Matched)
1	**OOC**	NewsCLIPpings (Mismatched)
2	**Manipulated**	DGM4 / MMFakeBench (Image Tampered + Real Text)
3	**AI-Text**	MMFakeBench (Real Image + AI-Text)
4	**Double Fake**	MMFakeBench (AI-Image + AI-Text)

3. UnivFD Stage: Forensic Image Path (Pre-training) Enhancement

Train the **UnivFD** model to become an expert in frequency artifacts before system integration.

3.1 DCT

The image is divided into patches before the DCT transformation.

1. **Patch Division:** Divide the 224 * 224 image into non-overlapping patches (8 * 8 or 16 * 16 blocks).
2. **Y-Channel:** Convert each patch to YCbCr and isolate the **Y-channel**.
3. **2D-DCT:** Apply the Discrete Cosine Transform to each block.
4. **Log Scaling:** Apply log(|DCT| + 1e-8) to compress the dynamic range.
5. **Normalization:** Min-Max scaling per image to [0, 1].
6. **Storage:** Pre-compute and save outputs as .pt files to avoid re-calculation during training.

3.2 Model & Training

- **model:** [UnivFD (ResNet-based)].
- Modify the first convolution layer (conv1) to accept **1 input channel** instead of 3.
- **Kaiming Normal Initialization:** Since the original conv1 weights are discarded, initialize the new layer's weights using **Kaiming Normal**. This ensures gradient stability at the start of training.
- Load pre-trained weights (blur_jpg_v0.pth) using strict=False to skip the modified first layer.
- **Loss:** BCEWithLogitsLoss.(binary)
- **Optimizer:** AdamW (Learning Rate: 1 * 10^{-4}).
- **Output:** Save as forensic_model.pth.

4. Textual Forensic Path (Fluoroscopy)

This module detects AI writing styles **by analyzing the internal representations** of LLM.

1. **Model Selection:** Use [Qwen2-7B-Instruct](#).
2. **Tokenization:**
   - Use AutoTokenizer from the transformers library.
   - Convert raw text into numerical **Input IDs** and **Attention Masks**.
3. **Hidden State Extraction:**
   - Perform a forward pass through the model with output_hidden_states=True.
   - **Layer 30:** Extract the tensor from the layer 30 transformer block. This layer contains the most relevant forensic features.
4. **Mean Pooling:**
   - The output is a matrix [Batch, Sequence, 4096].
   - Average the values across the sequence dimension to produce a single vector [Batch, 4096].
   - **MUST** apply the `attention_mask` before averaging to ensure that "padding tokens" do not dilute the forensic signal, preserving the integrity of the text embedding.
5. **Linear Projection:**
   - Pass the 4096-dim vector through a Linear (4096, 768) layer.
   - Use GELU as activation function for the projection.

5. Multimodal Fusion Architecture: Pairwise Interaction & 1D-Conv Orchestration

5.1 Layer Normalization

- Apply LayerNorm to v_semantic, v_imagFor, and v_textfor independently → **To** unify the mathematical space of heterogeneous signals coming from different models (FND-CLIP, UnivFD, Qwen), ensuring they are statistically comparable.

5.2 Pairwise Grouping (The 3 Interaction Channels) Organize the normalized vectors into three pairs for the cross-attention process:

- Pair 1 (Semantic <-> Image Forensic): Focuses on contextual-visual consistency.
- Pair 2 (Semantic <-> Text Forensic): Focuses on contextual-stylistic consistency.
- Pair 3 (Image Forensic <-> Text Forensic): Directly matches digital fingerprints.

5.3 Bidirectional Cross-Attention (The 6-Directional Inquiry)

- Perform a bidirectional attention mechanism for each pair. For any two vectors (x, y):
- Direction 1: x queries y (Q=x, K=y, V=y).
- Direction 2: y queries x (Q=y, K=x, V=x).
- Summation Logic: For each pair, instead of concatenating the two attention directions, **sum them element-wise (Output = Direction_1 + Direction_2).**
- Output: 3 relational vectors.
- Use **Multi-Head Attention (MHA) with 8 heads**. For each pair, calculate the attention twice: once where v_1 is the Query, and once where v_2 is the Query. This ensures the model captures multiple levels of forensic and semantic relationships.

5.4 1D-Convolution Fusion (Feature Integration)

Instead of simple concatenation, we use a **1D-Convolutional Layer** to process the merged outputs. **1D-Conv** acts as a filter that identifies correlations across the three interaction channels and automatically weights the most suspicious signals.

- Concatenate the 3 vectors into a matrix of shape (Batch, 3, 768).
- **Dimensional Permutation:** Apply x.permute(0, 2, 1) to transform the shape to (Batch, 768, 3). This ensures the convolution filter slides **across the 3 interaction channels** rather than inside a single vector.
- Apply 1D-Conv -> GELU -> 1D-Conv. The first layer extracts inter-pair correlations; the second acts as a bottleneck to compress forensic essence.
- **Output:** A compressed, high-density fused vector.

5.5 Training Configurations

Parameter	Value	Technical Detail
Batch Size	64	Balances gradient stability and GPU memory (VRAM).
Optimizer	AdamW	Learning Rate: 1 *10^{-4} with Weight Decay to prevent overfitting.
Total Loss	Loss_{Main} + (0.1 * Loss_{Aux})	Main Loss: CrossEntropy (5 classes). Aux Loss: BCE (Binary Forensic consistency) for UnivFD path. **Gradient Isolation:** Use detach() for the Auxiliary Loss to prevent forensic errors from corrupting the frozen LLM/Semantic backbones."
LR Scheduler	StepLR	Decay factor: 0.1 at Epoch 30.
Early Stopping	Patience = 10	Monitored on Validation F1-Macro.

6. Final Classification (MLP Classifier)

The output of the 1D-Conv enters a 3-layer MLP:

Layer 1: Linear(1024, 512) + BatchNorm + GELU + Dropout(0.5).

Layer 2: Linear(512, 256) + GELU

Layer 3: Linear(256, 5) → Produces 5 raw logits for the 5 classes.

***Dropout (0.5)** is essential to prevent the model from over-relying on a single modality, forcing it to look at the combined relational evidence.

7. Evaluation

1. Inference Mode: Deactivate Auxiliary head; use only Main Logits + Softmax.

   Ensure Softmax is applied only during inference, or within the loss function (CrossEntropyLoss) during training. The final layer must output raw logits for numerical stability.

2. Evaluate on the MMFakeBench Test Set to check for robustness against unknown AI generators.
3. Metrics: Precision, Recall, F1-Macro, and a single Confusion Matrix to analyze misclassification patterns